Five-Scale Audit of an e-5-137 Finite-Field Parameterization

I. Mikhailov

Independent Researcher, Moscow, Russia

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Abstract

This note records a finite-field parameterization built from e, π, α^{-1}, $N = 5$, $D = 26$, and arithmetic over GF(137). A formula atlas first lists standard reference equations from mechanics, electromagnetism, thermodynamics, relativity, quantum mechanics, nuclear physics, and cosmology. The project-specific material is then organized as a five-scale audit: MeV nuclear scales, eV lepton and QED scales, molecular and semantic repair scales, macroscopic gas and thermodynamic coarse-graining, and cosmological closure tests. The construction is phenomenological. It is not a replacement for the Standard Model, QCD, statistical mechanics, or ΛCDM. Each section therefore reports either a numerical benchmark or a null result.

1 Scope

The project uses a small set of integer and transcendental constants to define testable formulae:

$$N = 5, \qquad D = 26, \qquad p = 137. \tag{1}$$

The finite-field arithmetic uses residues in

$$\mathbb{F}_{137} \simeq \mathbb{Z}/137\mathbb{Z}. \tag{2}$$

The main dimensionless compression operator is

$$q = \frac{N(N-2)}{e^4\pi^3} = 0.008860611874290873. \tag{3}$$

The phase difference used in several audits is

$$\delta\phi = \cos(1/N) - \cos(2/N) = 0.05900558383835652. \tag{4}$$

The finite-field threshold used by the integer kernels and repair channels is

$$I_5 = 2(D - N) = 42. \tag{5}$$

The purpose of the note is not to infer physics from numerology. The purpose is to record which formulae survive fixed tests and which do not. A retained formula must specify its domain, inputs, residuals, and failure modes.

This paper belongs to a companion series of e-5-137 and GF(137) audit repositories. The repositories intentionally share notation, implementation patterns, and reproducibility conventions. The scope of this archive is the finite-field audit, the five-test matrix, and the English/Russian preprint artifacts specific to this paper; related repositories should be cited separately when their own results are used.

2 Common Operators and Data Structures

The computational layer uses three reusable objects.

First, a byte-valued residue vector stores model parameters:

$$x, w, b \in \mathbb{F}_{137}. \tag{6}$$

The edge-inference kernel computes

$$R = (XW + b) \bmod 137, \qquad A_{ij} = \mathbf{1}_{R_{ij} \geq 42}. \tag{7}$$

This is an integer inference primitive, not a floating-point neural-network approximation theorem.

Second, the fault-tolerance layer replicates a symbol s over 26 axes:

$$r_a = s + m_a(k) \pmod{137}, \qquad a = 0, \ldots, 25, \tag{8}$$

where $m_a(k)$ is a deterministic axis mask. Repair subtracts the mask and selects the most frequent residue:

$$\hat{s} = \arg\max_{v \in \mathbb{F}_{137}} \#\{a : r_a - m_a(k) \equiv v \pmod{137}\}. \tag{9}$$

With at most 10 corrupted axes, at least 16 correct axes remain:

$$26 - 10 = 16. \tag{10}$$

This gives a fixed repair budget for the toy channel.

Third, the long-context compression benchmark maps a 120,000 token object to

$$V_{\text{ref}} = 117 \tag{11}$$

semantic vectors, or

$$\frac{120{,}000}{117} = 1025.6410256410256 \quad \text{tokens/vector}. \tag{12}$$

This is a synthetic benchmark tied to the repository tests. It is not a general lossless-compression theorem.

3 Formula Atlas Across Physical Scales

This section lists the standard formulae used as reference structure from school mechanics through undergraduate field theory and cosmology. The finite-field parameterization is not used to replace these equations. It is used only as an audit layer when a discrete residue model, compression channel, or numerical benchmark is explicitly defined.

3.1 Dimensional and Conservation Prefilter

Every formula in the audit must first pass dimensional analysis:

$$[v] = LT^{-1}, \qquad [a] = LT^{-2}, \qquad [F] = MLT^{-2}, \qquad [E] = ML^2T^{-2}. \tag{13}$$

The conserved quantities used as organizing checks are

$$\frac{dE}{dt} = 0, \qquad \frac{dp}{dt} = 0, \qquad \frac{dL}{dt} = 0, \tag{14}$$

when the corresponding time-translation, space-translation, and rotational symmetries hold. A finite-field expression that violates dimensions is not retained as a physical formula.

3.2 Kinematics and Newtonian Mechanics

The kinematic definitions are

$$v = \frac{dx}{dt}, \qquad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}. \tag{15}$$

For constant acceleration,

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2, \tag{16}$$
$$v(t) = v_0 + at, \tag{17}$$
$$v^2 = v_0^2 + 2a(x - x_0). \tag{18}$$

Newtonian dynamics uses

$$F = ma, \qquad p = mv, \qquad J = \int F\,dt = \Delta p. \tag{19}$$

The work and energy formulae are

$$W = \int F\,dx, \qquad K = \frac{1}{2}mv^2, \qquad U_g = mgh, \qquad U_s = \frac{1}{2}kx^2. \tag{20}$$

The Lagrangian form is

$$L = T - V, \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0. \tag{21}$$

The e-5-137 layer has no role in these identities except as a possible integer representation of a numerical simulator.

3.3 Gravitation and Orbital Dynamics

Newtonian gravity is

$$F = G\frac{m_1 m_2}{r^2}, \qquad g = \frac{GM}{r^2}, \qquad \Phi = -\frac{GM}{r}. \tag{22}$$

For a circular orbit and escape from radius r,

$$v_{\mathrm{orb}} = \sqrt{\frac{GM}{r}}, \qquad v_{\mathrm{esc}} = \sqrt{\frac{2GM}{r}}. \tag{23}$$

Kepler's third law is

$$T^2 = \frac{4\pi^2}{GM}a^3. \tag{24}$$

The relativistic extension used later in the cosmology section is not derived from this Newtonian block; it is a separate metric theory.

3.4 Oscillations, Waves, and Optics

The harmonic oscillator obeys

$$m\ddot{x} + kx = 0, \qquad \omega = \sqrt{\frac{k}{m}}, \qquad T = \frac{2\pi}{\omega}. \tag{25}$$

The one-dimensional wave equation is

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}, \qquad v = \lambda f. \tag{26}$$

For geometric and wave optics,

$$n = \frac{c}{v}, \qquad n_1 \sin\theta_1 = n_2 \sin\theta_2, \tag{27}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}, \qquad m = -\frac{d_i}{d_o}, \tag{28}$$

and diffraction maxima satisfy

$$d \sin\theta = m\lambda. \tag{29}$$

These formulae provide the continuous wave reference for the discrete phase-lane language used by the repository.

3.5 Electricity and Magnetism

The electrostatic force and field definitions are

$$F = k_e \frac{q_1 q_2}{r^2}, \qquad E = \frac{F}{q}, \qquad V = \frac{W}{q}. \tag{30}$$

Circuit identities are

$$I = \frac{dq}{dt}, \qquad U = IR, \qquad P = UI = I^2 R = \frac{U^2}{R}. \tag{31}$$

For capacitors and inductors,

$$C = \frac{Q}{U}, \qquad E_C = \frac{1}{2} C U^2, \qquad E_L = \frac{1}{2} L I^2. \tag{32}$$

The Lorentz force is

$$F = q(E + v \times B). \tag{33}$$

Maxwell's equations in SI units are

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}, \qquad \nabla \cdot B = 0, \tag{34}$$

$$\nabla \times E = -\frac{\partial B}{\partial t}, \qquad \nabla \times B = \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial E}{\partial t}. \tag{35}$$

The fine-structure constant enters the lepton parameterization through α^{-1}, but the Maxwell equations themselves are not modified.

3.6 Thermodynamics and Statistical Mechanics

The ideal-gas equation is

$$PV = nRT = N k_B T. \tag{36}$$

The first and second laws are written as

$$\Delta U = Q - W, \qquad dS \geq \frac{\delta Q}{T}. \tag{37}$$

For heat capacity and phase transitions,

$$Q = mc\Delta T, \qquad Q = \lambda m. \tag{38}$$

The Carnot efficiency is

$$\eta = 1 - \frac{T_c}{T_h}. \tag{39}$$

The statistical form is

$$S = k_B \ln \Omega, \qquad p_i = \frac{e^{-\beta E_i}}{Z}, \qquad Z = \sum_i e^{-\beta E_i}. \tag{40}$$

In this paper, GF(137) appears at this level only as a finite microscopic alphabet whose randomized high-occupancy limit can be coarse-grained to the usual Boltzmann distribution.

3.7 Special Relativity

The Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{41}$$

Time dilation, length contraction, and energy-momentum relations are

$$\Delta t' = \gamma \Delta t, \qquad L' = \frac{L}{\gamma}, \tag{42}$$

$$E = mc^2, \qquad E^2 = p^2c^2 + m^2c^4. \tag{43}$$

The horizon-gated dark-energy audit uses a time-dilation-like factor, but its numerical effect is recorded as a null result in the cosmology block.

3.8 Quantum Mechanics

The Planck and de Broglie relations are

$$E = hf = \hbar\omega, \qquad p = \hbar k = \frac{h}{\lambda}. \tag{44}$$

The uncertainty relation is

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}. \tag{45}$$

The Schrödinger equation is

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi, \qquad \hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V. \tag{46}$$

For hydrogenic levels,

$$E_n = -13.6 \text{ eV} \frac{Z^2}{n^2}. \tag{47}$$

The project uses $5e$ eV = 13.591409142295225 eV as a numerical anchor near the hydrogen scale. This is a parameterization choice, not a derivation of QED.

3.9 Nuclear and Particle Bookkeeping

Nuclear binding and radioactive decay use

$$E_b = \Delta mc^2, \qquad R = R_0 A^{1/3}, \tag{48}$$

$$N(t) = N_0 e^{-\lambda t}, \qquad T_{1/2} = \frac{\ln 2}{\lambda}. \tag{49}$$

Reaction energy is

$$Q = (m_{\text{initial}} - m_{\text{final}})c^2. \tag{50}$$

The repository's nuclear calculation is tested against $E_b = \Delta mc^2$ and fails for helium, carbon, and iron at the < 10 ppm criterion.

3.10 Cosmological Observables

The Hubble relation and redshift definition are

$$v = H_0 d, \qquad 1 + z = \frac{a_0}{a}. \tag{51}$$

The critical density is

$$\rho_c = \frac{3H^2}{8\pi G}. \tag{52}$$

The Friedmann equation is

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}. \tag{53}$$

The equation-of-state parameter is

$$w = \frac{p}{\rho c^2}. \tag{54}$$

The effective radiation count is

$$N_{\text{eff}} = 3.046 + \Delta N_{\text{eff}}. \tag{55}$$

The model-specific CMB audit uses

$$\Delta N_{\text{eff}} = \frac{D}{137}\delta\phi(1 - q) = 0.011098917626279356. \tag{56}$$

3.11 Atlas Status Matrix

4 Five-Scale Physical and Computational Audit

4.1 Level 1: MeV Nuclear and QCD Scale

The MeV-scale audit tests whether the $D = 26$, 42, and 117 operators predict nuclear mass defects. The tested form is

$$\Delta M(Z, A) = Z^2 E_{\text{step}}(Z)\left[1 + \frac{42}{137D}\left(1 - \frac{13}{117}\right)\right], \tag{57}$$

where the running step is

$$E_{\text{step}}(1) = 5e \text{ eV}, \tag{58}$$

and

$$E_{\text{step}}(Z > 1) = 5e\,(137)(42)\left(1 - \frac{D}{137N}\right) \text{eV} = 0.07523660444808945 \text{ MeV}. \tag{59}$$

Table 1: Reference formula classes and the role of the finite-field layer.

Domain	Core formula	Conservation or limit	GF(137) role	Status
Mechanics	$F = ma$, $L = T - V$	energy, momentum, angular momentum	numerical representation only	reference theory
Gravity	$F = Gm_1m_2/r^2$	central-force orbits	none	reference theory
Waves/optics	$v = \lambda f$, Snell, lens law	phase propagation	phase-lane analogy	reference
EM	Maxwell equations, Lorentz force	charge and field continuity	α^{-1} in lepton ansatz only	reference
Thermal	$PV = nRT$, $S = k_B \ln \Omega$	maximum entropy limit	finite alphabet regulator	coarse-grained bridge
Quantum	$i\hbar\partial_t\psi = \hat{H}\psi$	unitary evolution	$5e$ eV anchor	parameterization
Nuclear	$E_b = \Delta mc^2$	mass-energy accounting	tested mass-defect encoder	null result
Cosmology	Friedmann, N_{eff}, w	homogeneous expansion	CMB and DESI audits	conditional test/null

The fixed correction factor is

$$1 + \frac{42}{137D}\left(1 - \frac{13}{117}\right) = 1.010481003181733. \tag{60}$$

Reference nuclear masses are obtained from neutral isotope masses after subtracting electron masses using CODATA constants [5, 6]. The audit table is:

Table 2: Nuclear mass-defect audit. The ppm residual is computed from the predicted nuclear mass relative to the tabulated nuclear mass.

Isotope	Z	Reference defect MeV	Model defect MeV	Residual ppm
^{1}H	1	0.0000133172	0.0000137339	−0.000444
^{4}He	2	28.2956897122	0.3041006382	7509.723753
^{12}C	6	92.1618167441	2.7369057434	8002.327108
^{56}Fe	26	492.259569845	51.3930078482	8463.590834

For iron,

$$Z_{\text{Fe}} = 26 = D, \tag{61}$$

but the numerical mass-defect gate fails:

$$\Delta M_{\text{Fe}}^{\text{model}} = 51.393007848156245 \text{ MeV}, \tag{62}$$

$$\Delta M_{\text{Fe}}^{\text{ref}} = 492.2595698447403 \text{ MeV}, \tag{63}$$

$$\epsilon_{\text{Fe}} = 8463.590833506674 \text{ ppm}. \tag{64}$$

The < 10 ppm criterion is not satisfied for helium, carbon, or iron. The MeV-scale result is therefore negative. The finite-field operators do not derive nuclear binding energies. QCD confinement, shell structure, pairing, and many-body nuclear dynamics remain outside this parameterization.

4.2 Level 2: eV/QED Lepton Scale

The eV-scale branch uses the electron mass as the normalization anchor:

$$m_e = 0.51099895069 \text{ MeV}, \qquad \alpha^{-1} = 137.035999084. \tag{65}$$

The base bookkeeping step is

$$5e \text{ eV} = 13.591409142295225 \text{ eV}. \tag{66}$$

The lepton mass ratios are written as generation-scale ansatz formulae:

$$\frac{m_\mu}{m_e} = \alpha^{-1}\left(\frac{3}{2} + q\right) = 206.768221426689, \tag{67}$$

and

$$\frac{m_\tau}{m_e} = \alpha^{-1}\left(N^2 + qf_\tau\right), \tag{68}$$

with

$$f_\tau = 42 + 2e^{-2} + 2e^{-7}. \tag{69}$$

The resulting tau/electron ratio is

$$3477.2282035579738. \tag{70}$$

The fourth charged benchmark is

$$\frac{m_{\tau'}}{m_e} = \alpha^{-1}\left[D^2 + q(DN)\right] = 92794.18434487357, \tag{71}$$

or

$$m_{\tau'} = 47.417730830364825 \text{ GeV}. \tag{72}$$

This is below current ordinary sequential charged-lepton limits [1]. It is therefore not an allowed Standard-Model-like fourth charged lepton.

The sterile neutral benchmark is

$$m_{\nu_{\tau'}} = m_e\left(\frac{F_{26}}{\alpha^{-1}}\right)\left[1 + \frac{DN}{\pi}\delta\phi(1-s)\right], \tag{73}$$

where

$$F_{26} = 121393, \qquad s = \frac{\alpha}{2\pi}. \tag{74}$$

The value is

$$m_{\nu_{\tau'}} = 1.5566463427697075 \text{ GeV}. \tag{75}$$

This is a mass benchmark only. Relic abundance, lifetime, production history, mixing angles, and detection limits require a separate particle model [2].

4.3 Level 3: Molecular, Repair, and Semantic Scale

Molecular binding energies lie in the eV range, but the repository does not contain an electronic-structure solver. The finite-field contribution at this scale is a symbolic repair and compression channel rather than a molecular Hamiltonian.

The threshold 42 is used as a discrete activation and repair marker:

$$A = \mathbf{1}_{R\geq 42}. \tag{76}$$

For the DNA toy channel, a symbol is encoded into 26 masked residues. If at most 10 axes are corrupted, the majority repair rule recovers the symbol in the tested model:

$$d_{\text{damaged}} \leq 10 \quad \Rightarrow \quad 26 - d_{\text{damaged}} \geq 16. \tag{77}$$

The test suite verifies this condition for synthetic DNA replication and for a semantic archive. In the semantic compression benchmark,

$$120{,}000 \text{ tokens} \mapsto 117 \text{ vectors} \mapsto 117 \times 130 \text{ bytes}. \tag{78}$$

The repair channel is exact under the configured damage budget. This is a statement about the test harness. It is not a proof of biological aging, organism-level longevity, or chemical stability.

4.4 Level 4: Macro/Gas Thermodynamic Limit

The macroscopic gas scale is treated as a coarse-graining limit of a finite alphabet. Let each microscopic degree of freedom carry a residue $a \in \mathbb{F}_{137}$. Map it to a centered integer

$$\rho(a) \in \{-68, -67, \ldots, 67, 68\}. \tag{79}$$

For a three-component velocity surrogate, use

$$v = (\rho(a_1), \rho(a_2), \rho(a_3))v_0, \tag{80}$$

and assign kinetic energy

$$\varepsilon(v) = \frac{m}{2}|v|^2. \tag{81}$$

For M independent cells, the maximum-entropy distribution under fixed mean energy is obtained by maximizing

$$S = -k_B \sum_i p_i \ln p_i \tag{82}$$

subject to

$$\sum_i p_i = 1, \qquad \sum_i p_i \varepsilon_i = \bar{E}. \tag{83}$$

The Lagrange multiplier calculation gives

$$p_i = \frac{e^{-\beta\varepsilon_i}}{Z_1(\beta)}, \qquad Z_1(\beta) = \sum_i e^{-\beta\varepsilon_i}. \tag{84}$$

Thus the discrete residue alphabet becomes a finite regularization of the Boltzmann distribution.

In the isotropic high-occupancy limit, the residue sum is replaced by a velocity integral. For N_g particles in volume V,

$$Z_{N_g} = \frac{1}{N_g!}\left[\frac{V}{h^3}\int_{\mathbb{R}^3} e^{-\beta p^2/(2m)}\, d^3p\right]^{N_g} = \frac{1}{N_g!}\left[V\left(\frac{2\pi m}{\beta h^2}\right)^{3/2}\right]^{N_g}. \tag{85}$$

The pressure follows from

$$P = \frac{1}{\beta}\frac{\partial \ln Z_{N_g}}{\partial V} = \frac{N_g}{\beta V}. \tag{86}$$

Since $\beta^{-1} = k_B T$,

$$PV = N_g k_B T = nRT. \tag{87}$$

The Boltzmann entropy is the corresponding macrostate count:

$$S = k_B \ln \Omega. \tag{88}$$

The result is a standard statistical-mechanics derivation written with a finite residue alphabet as the microscopic regulator. The ideal-gas law is not unique to GF(137). The role of GF(137) is to give a bounded discrete state space. When phase information is randomized and the distribution becomes isotropic, the modular order is washed out and the continuum thermodynamic limit recovers the Mendeleev–Clapeyron equation and Boltzmann entropy [8, 9].

4.5 Level 5: Cosmological Scale

The cosmological branch contains three tests.

First, the Hubble reference values used by the repository are

$$H_{0,\text{Planck}} = 67.4, \qquad H_{0,\text{SH0ES}} = 73.04, \tag{89}$$

with fractional difference

$$\frac{73.04 - 67.4}{67.4} = 0.0836795252225519. \tag{90}$$

The same section records a phenomenological CPL pair

$$w_0 = -0.7519948527423932, \qquad w_a = -0.8600649695978589. \tag{91}$$

Second, the Lagrangian dark-energy audit tested the two-field background

$$\mathcal{L}_{\text{dark}} = -\frac{1}{2}(\partial\phi)^2 + \frac{1}{2}(\partial\chi)^2 - \Lambda^4 U(x, y). \tag{92}$$

The horizon time-dilation gate was

$$H_{\text{gate}}(z, x) = \left[1 - \frac{q}{117} e^z \cos^2(5x)\right]^{-1/2}. \tag{93}$$

Its effect was too small:

$$\frac{q}{117} = 7.57317 \times 10^{-5}, \qquad \max H_{\text{gate}} \simeq 1.00028. \tag{94}$$

The accepted best row after the gate was

$$w_0 = -0.920331, \qquad w(z = 1) = -1.00259. \tag{95}$$

The target value was

$$w_{\text{target}}(z = 1) = -1.18202733754, \tag{96}$$

leaving

$$w(z = 1) - w_{\text{target}}(z = 1) = 0.179442. \tag{97}$$

The DESI-scale field-dynamics branch is therefore retained as an active null result, not as a fitted phantom-dark-energy model.

Third, the CMB closure invariant is

$$\Delta N_{\text{eff}} = \frac{D}{137}\delta\phi(1 - q) = 0.011098917626279356, \tag{98}$$

so that

$$N_{\text{eff}} = 3.046 + \Delta N_{\text{eff}} = 3.0570989176262793. \tag{99}$$

Relative to the Planck 2018 CMB+BAO value 2.99 ± 0.17 [3], this is

$$z_{\text{Planck}} = \frac{3.0570989176262793 - 2.99}{0.17} = 0.39469951544870036 \tag{100}$$

standard deviations from the central value. This is consistent with the absence of a large extra light free-streaming component in current extended CMB fits [4].

The CMB comparison also fixes the information-rate normalization used in the computational audit. The Planck-scale bit-rate density is treated as a bookkeeping scale,

$$\rho_{\dot{I},\mathrm{Pl}} \simeq 2.4 \times 10^{45}\ \mathrm{bits\ s^{-1}\ cm^{-3}}, \tag{101}$$

not as a measured throughput of the model. It is used only to express storage-bit reductions and finite-field update costs on a common scale.

For the sterile benchmark mass

$$m_{\nu_{\tau'}} = 1.5566463427697075\ \mathrm{GeV}, \tag{102}$$

the thermal-equivalent cutoff estimate

$$\lambda_{\mathrm{cut}} \simeq 0.11 \left(\frac{m}{\mathrm{keV}}\right)^{-4/3} \mathrm{Mpc} \tag{103}$$

gives

$$\lambda_{\mathrm{cut}} = 6.097325934119188 \times 10^{-7}\ \mathrm{kpc}. \tag{104}$$

This is far below the 100 kpc guardrail and far below DESI/BAO scales. The conclusion is conditional on a sterile, cold or sufficiently nonthermal production history.

5 Applied Integer Implementation

The edge-AI branch uses the same residue arithmetic for inference. The measured model sizes were

$$M_{\mathrm{GF}(137)} = 1.62793\ \mathrm{KB}, \qquad M_{\mathrm{float32}} = 6.50391\ \mathrm{KB}, \tag{105}$$

or a factor of approximately 4 in storage. For 1000 forward passes, the native ARM NEON loop took 12.1055 ms against 58.7784 ms for the NumPy float32 baseline:

$$\frac{58.7784}{12.1055} \simeq 4.86. \tag{106}$$

The call-through C++ path took 40.1530 ms, giving a speedup of about 1.46× while retaining byte-valued weights.

The release uses an energy-cost accounting estimate based on the Landauer lower bound for erasing B bits at temperature T,

$$E_{\mathrm{L}}(B,T) = Bk_{\mathrm{B}}T\ln 2. \tag{107}$$

For a stored state with $P = 2113$ field symbols, the FP32 representation uses

$$B_{\mathrm{FP32}} = 32P = 67616\ \mathrm{bits}. \tag{108}$$

The repaired finite-field state used in HYP-007 stores $R = 3458$ byte-valued symbols, hence

$$B_{\mathrm{RS}(26,16)} = 8R = 27664\ \mathrm{bits}, \qquad \frac{E_{\mathrm{L}}(B_{\mathrm{RS}},T)}{E_{\mathrm{L}}(B_{\mathrm{FP32}},T)} = 0.40913393279697113. \tag{109}$$

Across the released HYP-007 runs, the mean repaired-state ratio is

$$\left\langle \frac{E_{\mathrm{L}}(B_{\mathrm{RS}},T)}{E_{\mathrm{L}}(B_{\mathrm{FP32}},T)} \right\rangle = 0.4111824864372086, \tag{110}$$

Table 3: Edge-inference storage and runtime audit.

Backend	Memory KB	Time for 1000 passes	Role
float32 NumPy MLP	6.50391	58.7784 ms	host baseline
GF(137) C++ fused `uint8_t`	1.62793	40.1530 ms	call-through path
GF(137) C++ NEON native loop	1.62793	12.1055 ms	native loop

corresponding to a storage-bit accounting reduction of 0.5888175135627913. This is an accounting result, not a direct power measurement.

A hardware implementation would expose fixed operations

$$\texttt{mac137}(a, b, c) = (c + ab) \bmod 137, \tag{111}$$
$$\texttt{inv137}(a) = a^{-1} \bmod 137, \quad a \neq 0, \tag{112}$$
$$\texttt{ge42}(a) = \mathbf{1}_{a \geq 42}. \tag{113}$$

The relevant engineering property is bounded integer arithmetic. The `mac137` primitive can use Barrett reduction with

$$\mu = \left\lfloor \frac{2^{32}}{137} \right\rfloor = 31350126, \tag{114}$$

followed by one or more subtractive corrections. The `inv137` primitive is outside the hot inference loop and may be implemented as

$$\texttt{inv137}(a) = a^{135} \bmod 137 \tag{115}$$

by Fermat inversion or by a lookup table. A single primitive has bounded $O(1)$ cost; the two-layer dense audit kernel remains $O(RKH + RH)$ in the matrix dimensions, while a generic square dense GEMM is $O(N^3)$.

6 Summary Table

Table 4: Five-scale status summary.

Level	Scale	Main quantity	Status
1	MeV/QCD	Fe residual 8463.590833506674 ppm	failed < 10 ppm gate
2	eV/QED	$m_{\nu_{\tau'}} = 1.5566463427697075$ GeV	mass benchmark only
3	molecular/semantic	26 axes, 10-axis damage budget	toy repair passes
4	gas/thermal	$PV = N_g k_B T = nRT$	coarse-grained limit
5	cosmology	$N_{\text{eff}} = 3.0570989176262793$	within Planck window

7 Limitations

The failed nuclear mass-defect gate is part of the result. No correction term is added to force agreement with NIST/CODATA isotope masses. The lepton and sterile-neutrino formulae are parameterizations rather than a Lagrangian derivation. The molecular and DNA sections describe a symbolic finite-field repair code, not a biochemical model. The gas section is standard statistical mechanics with a finite residue alphabet used as a regulator. The cosmology section reports closure checks and null results under specified assumptions; it does not fit DESI, Planck, ACT, or Fermi-LAT likelihoods.

8 Conclusion

The e-5-137 construction is useful in this repository as a compact audit language. It gives reproducible integer kernels, fixed repair thresholds, several numerical mass benchmarks, and explicit negative results. The five-scale organization separates successful engineering tests from physics claims that remain unsupported. The retained scientific standard is the residual table, not the integer coincidence.

References

[1] S. Navas et al. (Particle Data Group), "Review of Particle Physics," *Phys. Rev. D* **110**, 030001 (2024), with 2025 update.

[2] R. Adhikari et al., "A White Paper on keV Sterile Neutrino Dark Matter," *JCAP* **01**, 025 (2017).

[3] N. Aghanim et al. (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209.

[4] Atacama Cosmology Telescope Collaboration, "The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models," arXiv:2503.14454.

[5] National Institute of Standards and Technology, "Atomic Weights and Isotopic Compositions with Relative Atomic Masses," `https://www.nist.gov/pml/atomic-weights-and-isotopic-compositions`.

[6] National Institute of Standards and Technology, "CODATA Recommended Values of the Fundamental Physical Constants: 2018," `https://pml.nist.gov/cuu/pdf/wall_2018.pdf`.

[7] Y. Kuratov, M. Arkhipov, A. Bulatov, and M. Burtsev, "Cramming 1568 Tokens into a Single Vector and Back Again: Exploring the Limits of Embedding Space Capacity," arXiv:2502.13063 (2025).

[8] L. Boltzmann, *Vorlesungen über Gastheorie*, J. A. Barth, Leipzig (1896).

[9] J. W. Gibbs, *Elementary Principles in Statistical Mechanics*, Charles Scribner's Sons, New York (1902).